

RESEARCH ARTICLE

Location estimation-based radio environment map construction in fading channels

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ABSTRACT

Latest regulations on TV white space communications and trend toward spectrum access through geolocation databases relax the regulatory constraints on cognitive radios. Radio environment map (REM) is a kind of improved geolocation database and an emerging topic with the latest regulations on TV white space communications. It constructs a comprehensive temperature map of the cognitive radio network operation area by utilizing multi-domain information from geolocation databases, characteristics of spectrum use, geographical terrain models, propagation environment, and regulations. REMs act as cognition engines by building long-term knowledge via processing spectrum measurements collected from sensors to estimate the state of locations without any measurement data. Active transmitter Location Estimation based REM construction technique is proposed and compared with the well-known REM construction techniques such as Kriging and inverse distance weighted interpolation in shadow and multipath fading channels. The simulation results suggest that the Location Estimation based REM construction outperforms the compared methods in terms of RMSE and correct detection zone ratio by utilizing additional information about channel parameters that can be estimated by classical least squares method easily. Copyright © 2013 John Wiley & Sons, Ltd.

KEYWORDS

cognitive radio; radio environment map; location estimation; correct detection zone ratio

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1. INTRODUCTION

Research on cognitive radio (CR) has matured from early works on the idea of CRs to today's multidimensional proposals such as signal processing techniques, machine learning algorithms, and regulations for cognitive radio. However, there still remains many technical challenges to overcome for evolving CRs into practical networks. TV White Space (TVWS), the portion of the ultra high frequency and very high frequency bands emerging after Digital Switch Over, has been an effective catalyst for the first practical applications of cognitive radio networks (CRNs). Initiated by the Federal Communications Commission (FCC) in USA, use of TVWS for dynamic spectrum access has attracted worldwide interest such as UK, Brazil, Japan, India, Singapore, and China. Digital Switch Over is almost completed in Japan except the three prefectures devastated by the 11th March earthquake, in Canada, some parts of Europe. It is scheduled to be completely finished in most of the Europe by 2012. South Korea, Australia, China, and Brazil are expected to switch off not later than 2015 [1].

White space measurements show that almost 50% of locations in UK have 150 MHz bandwidth, whereas 90% have 100 MHz capacity [2]. Similarly, measurement campaigns in many countries all over the world such as USA [3], Japan [4], and Spain indicate the potential of TVWS in meeting the escalating wireless connectivity requirements.

As the first step of opportunistic spectrum access, a TVWS device must establish the available frequencies at its location just before communications and must ensure the Primary Users (PUs) are not harmed. Due to the concerns mainly raised by the broadcasting industry on the proper protection of the incumbent TV services, regulatory bodies have brought strict rules on the white space devices. The rules demanded by the regulators, such as the detection of a primary signal at -114 dBm by FCC, are over-conservative. Such regulations reduce the available white spaces drastically (approximately by a factor of three) [3]. Instead, accessing a centralized database, which keeps track of the available spectrum based on geographical coordinates or the properties of registered transmitters, has been accepted more promising for mitigating

the strict sensing challenges while meeting the regulatory obligations.

Geolocation database, referred to as the TV band database, stores TV base station locations and the related parameters (static data) and constructs a real time view of the spectrum occupancy at the TV bands at each location [5]. Currently, database based TVWS operation is being pursued by regulators in the USA and the UK. In November 2009, FCC opened a call for database administrators and approved 10 companies (e.g., Google, Spectrum Bridge, Telcordia, etc.) as administrators in 2010. A step further from geolocation database lies the radio environment map (REM). REM was first defined as an abstraction of a real-world environment storing multi-domain information dynamically [6]. REM can act as an enabler for cognition in radios; it can store PU activity statistics as well as RF environment information such as propagation characteristics.

Essential functionality of an REM is constructing dynamic interference map for each frequency at each location of interest, which is volatile data. Because it is impractical to have measurements at each location in the CR operation area, REM fuses the available measurements to estimate the interference level at locations with no measurement data. Radio interference map interpolation is the most critical part of an REM construction method.

In this paper, we propose an active transmitter Location Estimation based (LIVE) REM construction technique in fading channels. We mainly focus on the radio interference map interpolation by utilizing the measurements from known locations. Proposed technique outperforms other REM construction techniques, and its time complexity is better than Kriging and inverse distance weighted interpolation, which are commonly used techniques.

LIVE uses received signal strength (RSS)-based active transmitter Location Estimation and enhances the performance of constructed REM. Due to the need for synchronization between primary and secondary networks, time-based algorithms such as time of arrival algorithm cannot be used in the CR systems. The problem that appears in angle of arrival is the lack of estimating the transmission power. Hence, among all of the common location estimation techniques, RSS-based algorithms render to be useful for simultaneous estimation of locations and powers of active transmitters in CR networks. If silent periods are employed in CR network, we can also differentiate the PU and the CR. Hence, we can also determine the type of the active transmitter.

RSS-based algorithms typically assume that the path loss exponent and correction are known. This is a frequently used assumption in the literature, because there are studies for estimating these channel parameters even without knowing exact locations of sensors [7]. In our case, we know the locations of measurement nodes. Hence, the process of estimating the path loss exponent and correction becomes classical least square regression analysis [8]. These measurement nodes periodically and in turn transmit at a predetermined power and known location that

enables the estimation of path loss exponent and correction accurately.

The rest of the paper is organized as follows. First, we give background information on geolocation databases and radio environment maps by providing details of REM data model and construction techniques in Section 2. Next, Section 3 gives details of the system model and the RSS measurements under fast and slow fading environments. Section 4 represents the performance evaluation of proposed method in terms of RMSE and correct detection zone ratio. Finally, Section 5 concludes the paper.

2. GEOLOCATION DATABASE AND RADIO ENVIRONMENT MAP

A geolocation database stores various information about PUs, and using this information it derives the available channel list at a location. CRs consult this database and retrieve the available channel list, including time constraints and maximum transmission power. Geolocation database keeps various information on PU (and possibly CR) transmitters: TV tower location, antenna height, user type (PU or CR), device transmit power, technology, operation channel(s), and duration of use [9].

REM was first defined as an abstraction of a real-world environment storing multi-domain information (e.g., PUs, policies, terrain data) [6]. However, it can also be considered more generally as an intelligent network entity that can further process the gathered information, inspect the spatio-temporal characteristics, and derive a map of the RF environment [10]. REM is a promising concept for efficient CRN operation without extensive burden on CRs as it can be considered as the cognitive engine located at the network. REM introduces environment awareness that would be harder to acquire by individual CR capabilities via extensive spectrum analysis. Hence, REM can also be seen as the network support turning simple nodes into intelligent CRs. The design of the REM relies on the following related processes: data gathering and representation, data processing/fusion, and data retrieval/query as depicted in Figure 1.

2.1. REM data model

Data stored in an REM can be classified into three as static, volatile, and derived [11]. Static data correspond to the information that does not change frequently, for example, the locations of TV towers, or the operators in a region. On the contrary, volatile data represent the information that is highly dynamic. CRs in a network and their spectrum use information can be categorized under this class. REM keeps its information about the environment up to date by dynamically tracking the changes. Lastly, derived data are interpreted from static and volatile data by some processing techniques, which is depicted in Figure 1.

Data stored in REM depend on the measurements collected from CRs in the network and other measurement

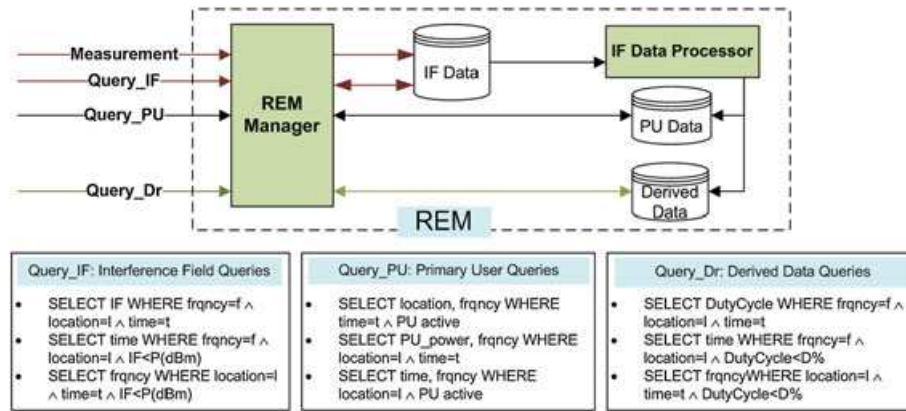


Figure 1. REM data flow and example queries.

nodes (a.k.a. measurement capable devices (MCDs)). Each measurement report also has the geolocation and time information. MCDs can be specific purpose sensor nodes, mobile phones, or similar devices.

2.2. REM construction techniques

REM construction is a wide concept referring to creating a complete map of CRN coverage area. However, we focus only on deriving interference level at each point of the CRN coverage area. This is also referred to as interference cartography (IC). IC forms a map of signal strength of RF environment as a function of spatial coordinates in a predefined area and, possibly, time (for a dynamic environment), and measurements from the CRs.

In the literature, REM construction techniques can be broadly put into two classes: spatial statistics based methods [12] and transmitter location determination based methods [13], which are also referred to as direct and indirect methods, respectively [11].

Spatial statistics describe the statistical properties of a given area utilizing the spatial correlational structure of this region. The fusion of the data from the examined area is commonly based on different types of interpolation methods. Interpolation is based on the basic principle that geographically closer locations are more related to each other compared with more distant locations. Common interpolation techniques for IC in the literature are Kriging, inverse distance weighted (IDW), and nearest neighbor interpolation. The frequency or volume of measurements, required accuracy in measurements, and density of measurement point are key factors determining the performance of each method. Although Kriging requires more measurement points, it is the most commonly applied technique in the literature [10,14] due to its higher precision.

If information on active transmitter locations are already available or can be approximated, they ease the

process of REM construction. Transmitter location determination based methods use this approach and first focus on localizing the transmitter(s) and deriving their properties. Subsequently, they estimate the signal strength at each location by applying the propagation modeling. However, this approach has more degrees of freedom: multiple transmitters, transmitter properties such as antenna propagation pattern, and accurate characterization of the propagation environment. In [12], Riihijärvi *et al.* study the effect of transmitter properties on the signal strength in a CRN using the second-order statistics. [13] applies an image processing based technique, which identifies the transmitters in the system and estimates their parameters based on RSS measurements from sensors. We follow this approach and first focus on estimating the active transmitter location and subsequently constructing the REM.

3. SYSTEM MODEL

We consider an environment in which the primary network reuses every frequency again in distant cells so that interference between primary users can be ignored. Thus, we consider a single active transmission for a single channel in the area of interest.

Our system model is depicted in Figure 2. MCDs are placed in a grid-like layout in the operation area, and a user is actively using a frequency band. These MCDs may be both devices deployed specifically by the CR operator or secondary user devices. Circular regions represent the different interference level contours. REM manager collects the (x, y, RSS) values from the measuring nodes and constructs the radio interference map via interpolating the RSS values at locations without any measurements.

RSS values depend on the characteristics of the propagation environment. We consider the log-normal shadowing and Rayleigh fading environments for fading channel analysis.

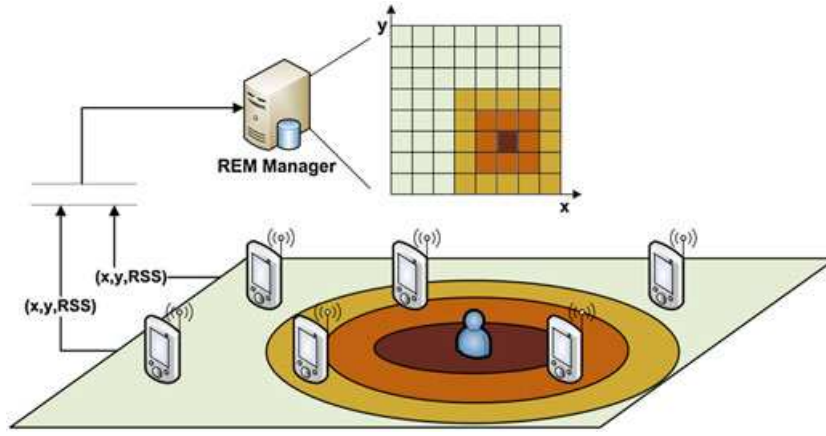


Figure 2. System model of CR operation area.

3.1. RSS measurements in slow fading channel

Long-term fading arises when the coherence time of the channel is large relative to the delay constraint of the channel. In this regime, the amplitude and phase change imposed by the channel can be considered roughly constant over the period of use. Slow fading can be caused by events such as shadowing, where a large obstruction such as a hill or large building obscures the main signal path between the transmitter and the receiver. The amplitude change caused by shadowing is often modeled using a log-normal distribution.

Under log-normal channel, RSS measurements in decibel obey normal distribution; hence, the analysis in decibel is more appropriate. The ideal RSS at the i th MCD, denoted by P_i^{rx} , is expressed as

$$P_i^{rx}[dB] = P^{tx}[dB] - P_{L_0} - 10 \log_{10} d_{(x_i, y_i)}^\alpha + S_i \quad (1)$$

where P_{L_0} and α are path loss correction and path loss exponents, respectively. $d_{(x_i, y_i)}$ is the distance between active user and the i th secondary user or MCD. S_i is a Gaussian random variable with mean zero and variance σ_s^2 for expressing the effect of log-normal shadowing.

RSS values at each MCD have severe disturbances caused by the shadowing effect, so we measure raw RSS values and evaluate sample mean to reduce the shadowing effect.

$$\overline{P_i^{rx}}[dB] = \sum_{j=1}^{N_m} P_{i,j}^{rx}[dB] / N_m \quad (2)$$

where $P_{i,j}^{rx}$ is the j th measured RSS value at the i th MCD and N_m is the total number of samples for unit measurement.

3.2. RSS measurements in fast fading channel

Short-term fading occurs when the coherence time of the channel is small relative to the delay constraint of the channel. In this regime, the amplitude and phase change imposed by the channel fluctuate considerably over the period of use. We use Rayleigh fading model over log-normal shadowing. Rayleigh fading is most applicable when there is no dominant propagation along a line of sight between the transmitter and receiver. In Rayleigh fading environment, instantaneous power exhibits an exponential distribution (not in dB). We consider shadowing/fading (log-normal shadowing plus Rayleigh fading) effects and use Suzuki distribution. The mobile channels modeled by Suzuki process already include multipath propagation and shadowing [15].

RSS measurements in watts obey exponential distribution with mean received from log-normal distribution; hence, the analysis in watts is more appropriate in this case. The ideal RSS at the i th MCD, denoted by P_i^{rx} , is expressed as

$$P_i^{rx}[Watts] = E_i \left(10^{\frac{\phi}{10}} \right) \quad (3)$$

where $E_i(\mu)$ is an exponential random variable with mean μ and

$$\phi = P^{tx}[dB] - P_{L_0} - 10 \log_{10} d_{(x_i, y_i)}^\alpha + S_i \quad (4)$$

where S_i is a Gaussian random variable with mean zero and variance σ_s^2 for expressing the effect of log-normal shadowing.

Because Rayleigh fading and shadowing cause statistically independent multiplicative fluctuations of the received power, attenuation in decibel values caused by shadowing and Rayleigh fading can be added. The results indicate that if logarithmic moments of the approximate log-normal distribution would be matched to the exact

Suzuki distribution, the mean would be found to be 2.5 dB below the local mean caused by shadowing [15]. Hence, to reduce the fading effects, we evaluate the sample mean of measurements and consider the loss due to short-term fading. Mean power in this case is

$$\overline{P_i^{rx}}[dB] = \sum_{j=1}^{N_m} P_{i,j}^{rx}[dB]/N_m + 2.5[dB] \quad (5)$$

where $P_{i,j}^{rx}$ is the j th measured RSS value at the i th MCD in watts and obeys exponential distribution.

3.3. Live REM construction

First, we estimate the active transmitter location and transmit power. After appropriate averaging, channel disturbance is minimized. If there is no disturbance in the measurements in decibels, then

$$10\alpha \log_{10} d_{(x_i, y_i)} \approx P^{tx} - P_{L_0} - \overline{P_i^{rx}} \quad (6)$$

Hence,

$$\sqrt{(x_{pu} - x_i)^2 + (y_{pu} - y_i)^2} \approx 10^{\frac{P^{tx} - P_{L_0} - \overline{P_i^{rx}}}{10\alpha}} \quad (7)$$

and after some manipulations we get

$$x_i^2 + y_i^2 \approx 2x_{pu}x_i + 2y_{pu}y_i + 10^{\frac{P^{tx} - P_{L_0} - \overline{P_i^{rx}}}{5\alpha}} - R^2 \quad (8)$$

where $R^2 = x_{pu}^2 + y_{pu}^2$, (x_i, y_i) and (x_{pu}, y_{pu}) are the locations of i th MCD and active user, respectively. Hence, we can express Equation (8) in the matrix form $A\theta \approx b$ where

$$A = \begin{pmatrix} 2x_1 & 2y_1 & 10^{\frac{-P_{L_0} - \overline{P_1^{rx}}}{5\alpha}} & -1 \\ 2x_2 & 2y_2 & 10^{\frac{-P_{L_0} - \overline{P_2^{rx}}}{5\alpha}} & -1 \\ \cdot & \cdot & \cdot & \cdot \\ 2x_{N_{MCD}} & 2y_{N_{MCD}} & 10^{\frac{-P_{L_0} - \overline{P_{N_{MCD}}^{rx}}}{5\alpha}} & -1 \end{pmatrix}$$

$$\theta = \begin{pmatrix} x_{pu} \\ y_{pu} \\ 10^{\frac{P^{tx}}{5\alpha}} \\ R^2 \end{pmatrix}, \quad b = \begin{pmatrix} x_1^2 + y_1^2 \\ x_2^2 + y_2^2 \\ \cdot \\ x_{N_{MCD}}^2 + y_{N_{MCD}}^2 \end{pmatrix}$$

The estimation problem formulated in the matrix form can be solved easily using least squares method [16,17]. We need at least four MCD measurements to solve for θ that has four unknowns. Let $\hat{\theta}$ be estimated value for θ . Then, the solution is computed as

$$\begin{aligned} \hat{\theta} &= \arg \min \|A\theta - b\| \\ &= (A^T A)^{-1} A^T b \end{aligned} \quad (9)$$

where $\|A\theta - b\| = (A\theta - b)^T (A\theta - b)$. Hence, we first estimate the location and the power of the active user.

Introducing nonlinear constraint $R^2 = x_{pu}^2 + y_{pu}^2$ results in

$$\begin{aligned} \hat{\theta} &= \arg \min \|A\theta - b\| \\ \text{s.t. } r^T \theta + \theta^T P \theta &= 0 \end{aligned} \quad (10)$$

where

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad r = \begin{pmatrix} 0 \\ 0 \\ 0 \\ -1 \end{pmatrix}$$

We solve the constrained optimization problem by transforming the original problem into Lagrange dual problem:

$$L(\theta, \lambda) = (A\theta - b)^T (A\theta - b) + \lambda(r^T \theta + \theta^T P \theta) \quad (11)$$

where λ is the Lagrange multiplier. We differentiate the Lagrangian with respect to θ and λ and get

$$\frac{\partial L}{\partial \theta} = 2\theta^T (A^T A + \lambda P) - 2b^T A + \lambda r^T \quad (12)$$

$$\frac{\partial L}{\partial \lambda} = r^T \theta + \theta^T P \theta \quad (13)$$

To satisfy the optimality condition, Equations (12) and (13) must be zero. Setting Equation (12) equal to zero leads to

$$\theta = (A^T A + \lambda P)^{-1} \left(A^T b - \frac{\lambda}{2} r \right) \quad (14)$$

Setting Equation (13) equal to zero and substituting θ found in Equation (14) results in single unknown variable equation in terms of λ only, which can be easily solved by numerical methods. After finding λ , we solve the optimization problem in Equation (10) by solving the Lagrange dual problem in Equation (11) using least squares method [18].

After solving for $\hat{\theta}$, we have the estimated values for (x_{pu}, y_{pu}) and P^{tx} . We proceed with the REM construction method by utilizing these estimated values $(\hat{x}_{pu}, \hat{y}_{pu})$ and $\hat{P}^{tx}$. We evaluate the estimated power at any location in Equation (15) by using the estimated data and channel information.

$$\overline{P^{rx}}(x, y)[dB] = \hat{P}^{tx}[dB] - P_{L_0} - 10\alpha \log_{10} \hat{d}_{(x, y)} \quad (15)$$

$$\text{where } \hat{d}_{(x, y)} = \sqrt{(x - \hat{x}_{pu})^2 + (y - \hat{y}_{pu})^2}.$$

3.4. REM quality

By utilizing the RSS measurements from all MCDs, the essential part of REM, the estimated interference map, is constructed at the network. However, estimated REM

and the actual REM may differ at some locations. How accurate an REM describes the real operation environment is measured by REM quality metrics. These quality metrics can be evaluated by comparing estimated REM with true REM in terms of correct detection zone, false alarm zone, and root-mean-square error (RMSE). Number of sensors, distribution of sensors in the field, capability of sensors, dynamics of the propagation environment, and accuracy of the propagation modeling are the key parameters affecting the performance of the REM construction method.

The intersection of the zones of true REM and estimated REM defines the zones that are correctly and incorrectly determined. In Figure 3, the true REM and the estimated REM contours are depicted. The solid line divides the area into two subareas in which the transmission is banned (no-talk-zone) and allowed (talk-zone), denoted by dark and light shaded areas. The subscripts of the zones depict the actual state with respect to true REM. The areas where the true REM and the estimated REM contours both determine H_1 or H_0 zones correctly are called the correct detection zone type-1 (Z_1^{CD}) and correct detection zone type-0 (Z_0^{CD}), respectively. False alarm zone (Z_0^{FA}) is the area in which transmission is forbidden because of estimation errors but the area is in fact allowed zone for transmission. Finally, missed detection zone (Z_1^{MD}) is the area where transmission is not allowed, but estimated REM infers the opposite. We normalize the regions according to true REM and propose the Z_1^{CD} and Z_0^{FA} ratios (CDZR₁ and FAZR), which present a better understanding about the performance of an REM algorithm;

$$\text{FAZR} = \frac{A(Z_0^{FA})}{A(Z_0^{FA}) + A(Z_0^{CD})}$$

$$\text{CDZR}_1 = \frac{A(Z_1^{CD})}{A(Z_1^{CD}) + A(Z_1^{MD})}$$

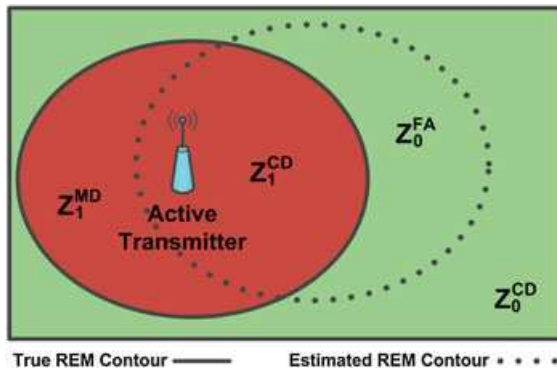


Figure 3. False alarm, missed detection, and correct detection zones.

where $A(Z)$ represents the area of the zone Z . These proposed REM performance metrics are analogous to probability of false alarm and detection. Hence, we can plot the ROC curves similar to detection context.

4. RESULTS

We consider an area of $1000\text{ m} \times 1000\text{ m}$ divided into subregions. In each subregion, an MCD at the center measures the average RSS and reports it to the REM manager. Path loss exponent, path loss correction, and σ_s are selected as 3.5, 38.4, and 8, respectively for an urban area [11]. IDW1 and IDW2 use inverse distance and inverse distance square weighted interpolation techniques.

4.1. Effect of channel conditions

In this subsection, we consider 16 MCDs (4×4 subregions with edge size $250\text{ m} \times 250\text{ m}$) in an area with log-normal shadowing and shadowing/fading channel conditions and compare the methods in terms of RMSE and ROC curves.

For analyzing the performance under different channels, we consider log-normal shadowing and combined shadowing/fading (log-normal shadowing plus Rayleigh fading) channel conditions. First, we compare the methods in terms of RMSE with respect to true REM and smoothed REM.

In all three conditions, proposed method achieves better performance than other methods because RMSE shows the deviation of the constructed REM from true REM. In Figures 4(a) and 4(b), RMSE values of the REM construction methods are depicted under log-normal shadowing and log-normal shadowing plus Rayleigh fading channel conditions, respectively. Under shadowing/fading channel, Kriging and L1vE REM construction methods are affected adversely. However, increasing the N_m (averaging sample size) decreases the adverse effects by eliminating the fading issues.

More useful performance metrics such as CDZR₁ and FAZR provide a better understanding of the network. In Figure 3, zones for true REM and estimated REM are depicted. Better REM construction methods fit better on the true REM contour with predetermined threshold value (-120 dB in our case). In Figure 5, FAZR versus CDZR₁ is depicted for understanding the achievable CDZR₁ for a given FAZR. Having higher CDZR₁ means disturbing the active user is less probable. Having less FAZR means, white spaces can be utilized better.

In Figure 5, L1vE REM outperforms other methods with the help of channel knowledge, which is a reasonable assumption. Knowing the channel parameters of the operating area such as path loss exponent and correction enables us to estimate primary user location and transmit power. Therefore, constructing the REM becomes easy with the help of the estimated data.

In Figure 5(a) and 5(b), ROC curves of the REM construction methods are depicted under log-normal shadowing and log-normal shadowing plus Rayleigh fading

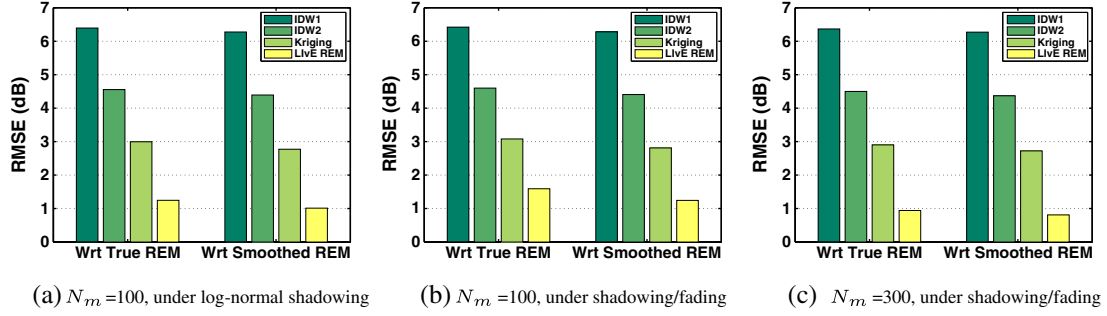


Figure 4. RMSE values for REM construction techniques.

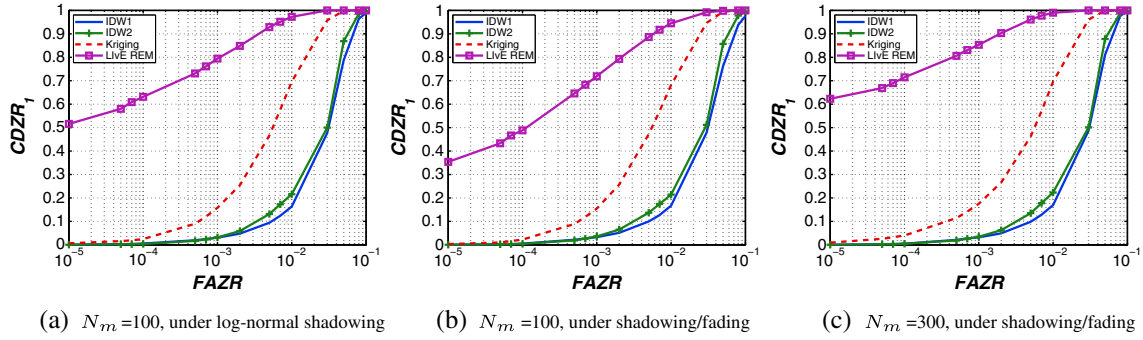


Figure 5. ROC curves of REM construction techniques.

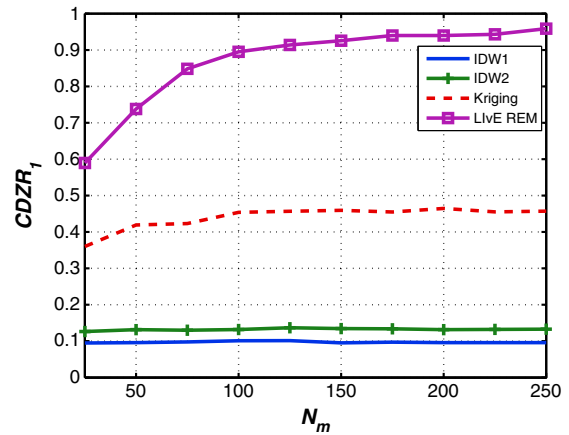
channel conditions, respectively. Under shadowing/fading channel Kriging and LiVE REM construction methods are affected adversely as in RMSE case. In Figure 5(c) for FAZR = 0.002, we have CDZR₁ values equal to 0.9, 0.026 and below for LiVE, Kriging and IDW1 and IDW2 REM construction methods, respectively. LiVE REM construction technique clearly outperforms the other methods. If we have CDZR₁ = 0.9 constraint, then achievable FAZR values are 0.002, 0.021, 0.062, and 0.065 for LiVE, Kriging, IDW1 and IDW2, respectively. Having less FAZR provides us better utilization of white spaces.

4.2. Effect of N_m on CDZR₁

Increasing the sample size for averaging RSS values mitigates the adverse effects of fading channel. Hence, it provides better statistics for estimating the active transmitter location and transmit power. If the channel conditions are harsher, then increasing the N_m value improves the performance of the LiVE REM construction method.

In Figure 6, N_m versus CDZR₁ is depicted with FAZR = 0.005 constraint and under shadowing/fading channel condition. Under these conditions, LiVE REM construction method outperforms the others. Only LiVE REM construction method can achieve acceptable CDZR₁ values.

Increasing N_m (up to a point) improves the results significantly for the Kriging and LiVE REM construction methods. Under shadowing/fading for LiVE REM method having $N_m > 200$ does not improve the system performance significantly, hence appropriate averaging sample size for RSS can be close to 200.

Figure 6. N_m versus CDZR₁ (FAZR=0.005, shadowing/fading).

4.3. Effect of distance to closest MCD on $CDZR_1$

In Figure 7, d_{min} versus $CDZR_1$ is depicted where d_{min} is the distance between active user and the closest MCD. Some interpolation techniques heavily depend on the location of active transmitter. If active user is close to any one of the MCDs, the resulting REM is very close to actual REM. However, this condition is not necessarily satisfied in real life. LiVE REM construction method does not depend on the location of the active user or closeness to any one of the MCDs. LiVE REM construction technique first estimates the active transmitter location and transmit power to remove the dependency.

Increasing d_{min} severely affects the performance of the Kriging, IDW1, and IDW2. LiVE REM construction technique is not affected by the d_{min} parameter. Therefore, LiVE REM is a robust and efficient REM construction technique that outperforms Kriging, IDW1, and IDW2.

4.4. Effect of N_s on $CDZR_1$

In Figure 8, N_s versus $CDZR_1$ is depicted under shadowing/fading. Because we divide the area into subregions in each direction equally and to make fair comparison, we consider perfect squares as the number of MCDs. LiVE REM performs better than other techniques in terms of $CDZR_1$ for each of the cases. Increasing N_s has a monotonic increasing effect on Kriging and LiVE REM. We observe that LiVE REM converges quickly a ratio of 0.7 for 9 MCDs to almost 1 for 16 MCDs. On the other hand, Kriging exhibits an unacceptable performance, a $CDZR_1$ of less than 0.1, for 9 MCDs and jumps to 0.58 for 16 MCDs and 0.75 for 25 MCDs. Even with this significant increase, Kriging is well below that of LiVE REM. The significant increase in Kriging can be explained by the

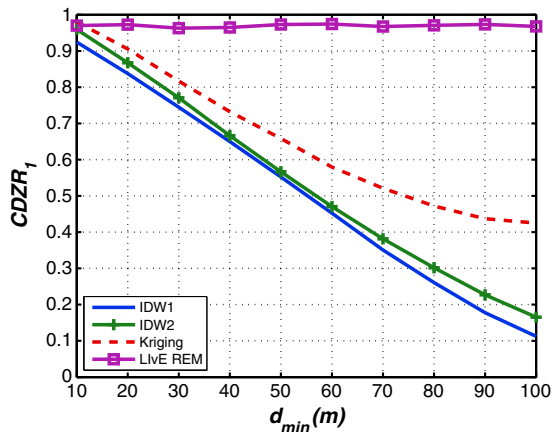


Figure 7. d_{min} versus $CDZR_1$ ($N_m = 300$, FAZR=0.005, under shadowing/fading).

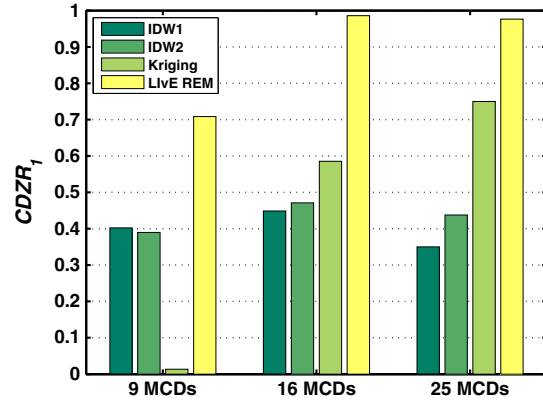


Figure 8. N_s versus $CDZR_1$ ($N_m = 300$, FAZR=0.005, under shadowing/fading).

fact that Kriging requires more sensors to work effectively [11]. However, our proposed method works with 9 MCDs effectively, and LiVE REM exhibits a better performance than all of the other methods even in the case of 9 MCDs. We observe that the performance of IDW1 and IDW2 are not significantly improved by the number of MCDs, but it fluctuates. This behavior can be explained by the fact that increasing number of MCDs changes the d_{min} adversely for some cases and these methods are close to their limits for the given parameters. By the help of Figure 8, a system designer can find the sufficient number of sensors to be placed for effective REM operation.

5. CONCLUSIONS

REMs act as cognition engines by building long-term knowledge via processing spectrum measurements collected from sensors to estimate the state of locations, which do not have any measurement data. LiVE REM construction technique is proposed and compared with the well-known REM construction techniques such as Kriging and inverse distance weighted interpolation under shadowing and multipath fading channels. The simulation results suggest that the location estimation based REM construction outperforms the compared methods in terms of RMSE and $CDZR_1$ by utilizing additional information of channel parameters, which is a reasonable assumption. We analyze the effect of N_m and d_{min} on $CDZR_1$ and conclude that increasing N_m mitigates the fading effects and improves the performance of LiVE REM significantly. Analysis on effect of d_{min} focuses on how the algorithms are affected by the location or closeness of active user to any MCDs. If active user is not close enough to any MCDs, the achievable $CDZR_1$ for Kriging for a given FAZR constraint is not reasonable ($d_{min} = 50$ implies $CDZR_1 \approx 0.65$). However, LiVE REM is not affected by the d_{min} . It always achieves $CDZR_1$ higher than 0.9 because it first estimates the location and the transmit power of active transmitter.

For the future directions, we analyze the LiVE REM performance in heterogeneous environments by utilizing mixed path loss formulations.

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